

A380 takeoff calculator and company T.O data

Flypad takeoff performance for the A380X, with the FMS company takeoff data exchange of the A380 FCOM

Branch	feature/a380/efb/take-off-calculator
Aircraft	A380X (flypad shared with the A32NX, see document 12)
Commits	c16e6e877 feat(a380x/efb): A380 takeoff calculator with company T.O data to the FMS
Base	feature/a380/mcdi-improvements (document 10)
Status	Merged in master (cherry-picked as dba687e4d). Tested in the sim by the user.

Purpose

The A380X flypad had no takeoff calculator. The A32NX one uses FBW's A320 performance model, which does not exist for the A380. This feature adds an A380 calculator built from public Airbus data and the FCOM rules, a runway drawing that shows the distance needed on the runway, and the FCOM company takeoff data function that sends the results to the MFD, where the crew inserts them on PERF T.O.

Specification

Data sources

- **Airbus A380 Aircraft Characteristics** (AC, 1 Dec 2025), take-off weight limitation charts, PDF p.135-139 (ISA and ISA+15, Trent 900 and GP7200). The curves are vector paths: they were extracted exactly (not digitised from pictures): runway 1500-5500 m, TOW 300-650 t, airport altitudes 0/2000/4000/6000/8000 ft. FBW's A380X has Trent 972 engines, so the Trent 900 charts are used. The charts are "for information only": dry runway, no wind, no slope.
- Tables in `a380x_takeoff_data.ts`: 1700-5100 m x 0-8000 ft, ISA and ISA+15. Below 1700 m the chart curves start distorted: the table does not go lower, shorter runways are extrapolated (W proportional to L).
- A380 FCOM for the rules (table below). The FBW VMCA / VMCG tables are A320 placeholders: the FCOM gives no values.

Two data modes

- **Airbus data + estimates** (default): chart weights, plus FBW-model estimates for what the charts do not cover (wind, slope, temperatures between the ISA and ISA+15 curves, FLEX). Estimated values are amber with an asterisk, as on the OIS.
- **Airbus data only**: only what the charts give; anything else is an error message instead of an estimate.

Model

- MTOW(perf) from the chart for the runway length and airport altitude, corrected for temperature: $W \sim \sigma^k \times \text{thrust}^j$, with k fitted on ISA vs ISA+15 (0.45 short runway to 0.2 long) and j on the altitude curves (0.3-0.45).
- FLEX: $W(T) = W(\text{TREF}) \times (\text{density ratio})^k \times \text{thrust ratio}$, thrust ratio linear from 1 at TREF to 0.6 at TMAXFLEX (FCOM: 40 % maximum reduction). Checked on the FCOM example (560 t on a 614 t MTOW -> FLEX 38 degC; model 37).
- Speeds: $V_2 \geq 1.13 V_{S1G}$ and $\geq 1.10 V_{MCA}$, $V_R \geq 1.05 V_{MCA}$, $V_1 \geq V_{MCG}$; V_R/V_2 and V_1/V_R ratios from the FCOM RESULTS example (V_1 153 / V_R 168 / V_2 174 at 560 t).
- Wind: regulatory 50 % of the headwind, 150 % of the tailwind. Line-up allowances 172.5 / 219.8 ft (90 / 180 deg entry).
- **Takeoff run on the runway**: the required length is the runway length at which the chart weight limit equals the TOW (bisection). V_1 , V_R and 35 ft points: $\text{TOD0} = \text{required} / 1.15$ (CS 25.113), constant acceleration to $V_2 + 10$.

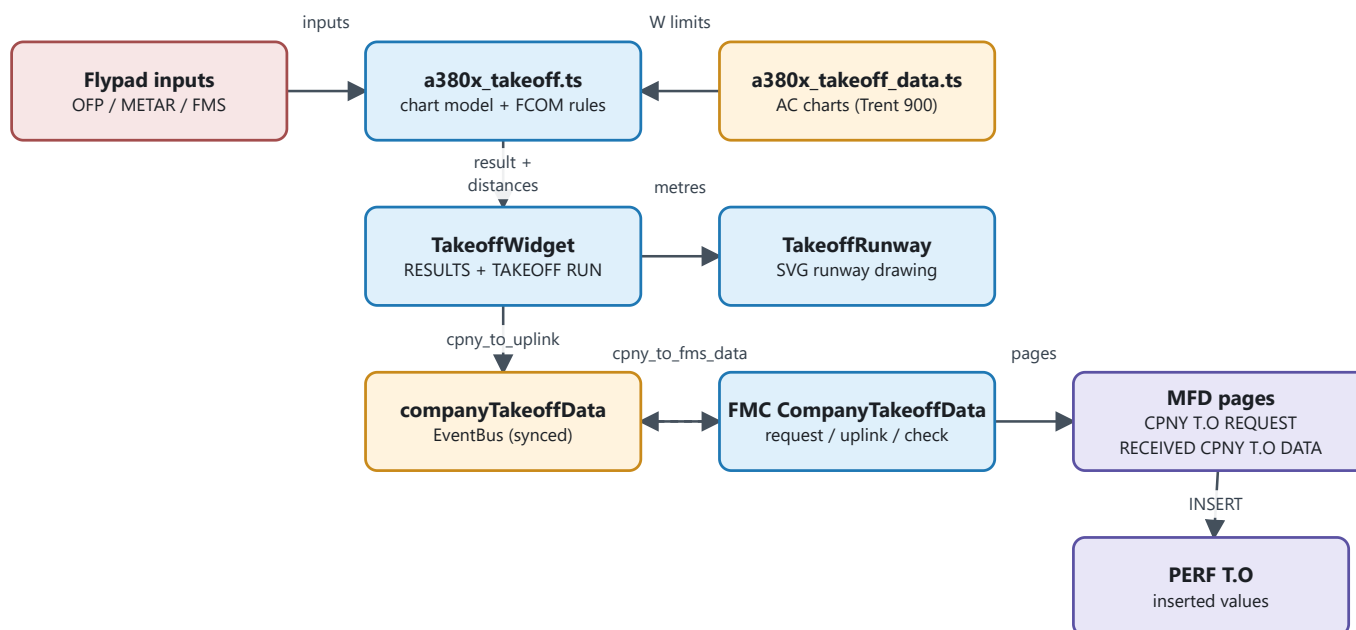
Screen

- Four input sections: RUNWAY (ident or manual TORA, heading, elevation, slope, entry angle), CONDITIONS (runway condition, wind, OAT, QNH, anti-ice, packs), AIRCRAFT (TOW, T.O CG, flaps, thrust), DEPARTURE (THR RED / ACCEL /

EO ACCEL, defaulted to the elevation + the flypad CONFIG altitudes; noise abatement).

- Fill from: SimBrief OFP, METAR, or **FMS** (the MFD answers with its takeoff data, including the runway conditions of the last SEND T.O REQUEST).
- RESULTS panel in the layout of the OIS takeoff application (FCOM PER-TOF-TOR-SRS P 1 figure): TOW, MTOW(perf), V1/VR/V2, margin, FLEX or TOGA, flaps, T.O CG, T.O SHIFT, THR RED / ACCEL / EO ACCEL.
- TAKEOFF RUN panel: runway drawing (length, shift, line-up, V1 / VR / 35 ft marks, required distance bar, margin), TOGA or any FLEX from the lowest allowed to the maximum with +/- buttons and a slider, whatever thrust was chosen first (FCOM: a lower FLEX may be used to increase the margin).

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture: calculator, flypad screen and the company T.O data path to the MFD

FCOM rules implemented

Rule	Source	Implementation
TREF = ISA+15, TMAXFLEX = ISA+60, max 40 % thrust reduction, no FLEX on a contaminated runway	FCOM PER-TOF-THR-FLX	FLEX range and thrust ratio; TOGA forced on contaminated runways
Range of TFLEX: any FLEX from max(TREF, OAT) to the maximum may be used	FCOM PER-TOF-THR-FLX	FLEX slider in the TAKEOFF RUN panel
V2 >= 1.10 VMCA, >= 1.13 VS1G; VR >= V1, >= 1.05 VMCA; V1 >= VMCG	FCOM PER-TOF-TOR-SRS P 1-2	speed limits
RESULTS panel layout	FCOM PER-TOF-TOR-SRS P 1 (PDF p.6609)	ResultsPanel
Uplink with or without request; message COMPANY T.O DATA RECEIVED WAITING FOR INSERTION	FCOM DSC-22-FMS-20-30	FMC CompanyTakeoffData
Up to 4 runways x 2 thrust ratings received	FCOM DSC-22-FMS-20-30	uplink list
Runway amber when not the active one; INSERT only with the active runway	FCOM DSC-22-FMS-20-30 P 312-320	RECEIVED page + check

TOW margin -2 t / +7 t of FUEL&LOAD, else amber + UPLINK/ACTIVE TOW DISAGREE	FCOM DSC-22-FMS-20-30	check before INSERT
INSERT needs speeds, TOW, T.O CG, wind, runway condition, baro, temperature	FCOM DSC-22-FMS-20-30	INSERT enabling
INSERT puts speeds, flaps, thrust, shift, THR RED / ACCEL / EO ACCEL, noise (not the CG: the crew enters it in THS FOR)	FCOM + SOP	insertCompanyTakeoffData
T.O LIMIT = remaining runway length (FT), T.O SHIFT in M; TEMP dashes for FLEX/DERATED	FCOM DSC-22-FMS-20-30	RECEIVED page fields
SEND T.O REQUEST -> REQUEST PENDING...; no answer in 4 min -> NO COMPANY REPLY	FCOM DSC-22-FMS-20-30	request timer
Required takeoff distance = 1.15 x TOD0 (all engines)	CS 25.113	V1 / VR / 35 ft marks

Interfaces

Item	Type	Meaning
cpny_to_fms_data	EventBus (synced)	FMS takeoff data / request, answers a flypad import
cpny_to_fms_data_request	EventBus	flypad asks the FMS for its data
cpny_to_uplink	EventBus	takeoff data sent by the flypad (one per runway and thrust)

Verification

- MFD and flypad checked in a local browser harness (received page fields, INSERT populating PERF T.O).
- Model checked against the FCOM example; unit-level checks in a Node harness.
- User in-sim test: accepted.

Limitations

- The charts are dry / no wind / no slope: every correction is an FBW estimate, flagged amber.
- VMCA / VMCG are the FBW A320 placeholders (no A380 values in the FCOM).
- Below 1700 m the required length is extrapolated (real-data mode: "less than 1700 m").

PR description (from the PR draft)

From `pr-drafts/11-a380-takeoff-calculator.md`, section "Summary of Changes" onwards.

Summary of Changes

The A380X flypad gets a takeoff calculator, and the MFD the FCOM company takeoff data function that receives its results.

Calculator (`a380x_takeoff.ts`, `a380x_takeoff_data.ts`)

- Weight limits from the Airbus A380 Aircraft Characteristics (1 Dec 2025), take-off weight limitation charts, Trent 900, ISA and ISA+15 (PDF p.135-139). The curves are vector paths and were extracted exactly: runway 1700-5100 m, airport altitude 0-8000 ft.
- Corrections the charts do not give are FBW-model estimates, amber with an asterisk: temperature ($W \sim \sigma^k \times \text{thrust}^j$, fitted on the charts), FLEX (thrust ratio 1 at TREF to 0.6 at TMAXFLEX, FCOM 40 % maximum; checked on the FCOM example 560 t / 614 t -> FLEX 38), wind (50 % headwind / 150 % tailwind), slope, line-up (172.5 / 219.8 ft). A "Data" select switches to Airbus data only.
- Speeds: $V2 \geq 1.13 VS1G$ and $\geq 1.10 VMCA$, $VR \geq 1.05 VMCA$, $V1 \geq VMCG$, ratios from the FCOM RESULTS example.
- Takeoff run: the required length is where the chart limit equals the TOW (bisection); $V1 / VR / 35$ ft points from $TOD0 = \text{required} / 1.15$ (CS 25.113) with a constant acceleration to $V2 + 10$.

Flypad screen: RUNWAY / CONDITIONS / AIRCRAFT / DEPARTURE inputs (fill from the OFP, the METAR or the FMS), the RESULTS panel of the OIS takeoff application (FCOM PER-TOF-TOR-SRS P 1), and a TAKEOFF RUN panel with a runway drawing (shift, line-up, $V1 / VR / 35$ ft marks, required distance, margin) for TOGA or any FLEX of the allowed range (FCOM PER-TOF-THR-FLX, range of TFLEX).

Company T.O data (FCOM DSC-22-FMS-20-30): SEND T.O REQUEST (REQUEST PENDING..., NO COMPANY REPLY after 4 min), uplinks for up to 4 runways x 2 thrust ratings, message COMPANY T.O DATA RECEIVED WAITING FOR INSERTION, INIT / PERF button RECEIVED CPNY T.O, the RECEIVED COMPANY T.O DATA page (runway amber when not the active one, TOW amber outside -2 t / +7 t with UPLINK/ACTIVE TOW DISAGREE, TEMP dashes for FLEX, T.O LIMIT and T.O SHIFT) and INSERT, which fills PERF T.O (speeds, flaps, thrust, shift, THR RED / ACCEL / EO ACCEL, noise; not the CG, which the crew enters in THS FOR).

Cockpit API Changes

None (EventBus topics `cpny_to_fms_data`, `cpny_to_fms_data_request`, `cpny_to_uplink`, synced between the instruments).

Screenshots (if necessary)

TO ADD: the flypad screen (TOGA and FLEX runs), the RECEIVED COMPANY T.O DATA page, PERF T.O after INSERT.

References

- Airbus A380 Aircraft Characteristics, Airport and Maintenance Planning (1 Dec 2025), take-off weight limitation charts.
- A380 FCOM PER-TOF-THR-FLX, PER-TOF-TOR-SRS P 1-2, DSC-22-FMS-20-30 P 312-320 (RECEIVED COMPANY T.O DATA). CS 25.113.

Additional context

The FBW VMCA / VMCG tables are A320 placeholders (the FCOM gives no values). The charts are dry, no wind, no slope: every other case is an estimate and flagged as one.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Fill from the OFP, calculate: results and takeoff run; switch TOGA / FLEX and move the FLEX slider: the distances change.
2. Data select "Airbus data only": corrections not covered by the charts give an error instead of an estimate.
3. SEND TO FMS, then MFD RECEIVED CPNY T.O: the data per runway / thrust; INSERT fills PERF T.O.
4. Change the MFD runway or the ZFW: the runway / TOW turns amber and INSERT is not available.
5. MFD SEND T.O REQUEST, then Fill from FMS in the flypad: the request conditions come in.